APT Regression Methods

Estimating Factor Returns and Risk Premia by Cross-Sectional Regression

IEOR E4576 — Data-Driven Models in Finance — Fall 2026 — Week 8

1 The Fundamental Factor Model

The Arbitrage Pricing Theory (APT) relates asset returns to a small number of systematic factors. In a **fundamental** factor model the exposures are *observed* characteristics of each stock — value (B/M, E/P), size, momentum, profitability, industry membership — and the factor returns are the unknowns to be estimated. For a universe of n assets and K factors,

$$\mathbf{r} = \mathbf{B}\,\mathbf{f} + \boldsymbol{\epsilon}, \tag{1}$$

where

- $\mathbf{r}$ is the $n \times 1$ vector of asset returns,
- $\mathbf{B}$ is the $n \times K$ matrix of factor exposures (row i holds the characteristics of asset i),
- $\mathbf{f}$ is the $K \times 1$ vector of factor returns,
- $\boldsymbol{\epsilon}$ is the $n \times 1$ vector of idiosyncratic (specific) returns.

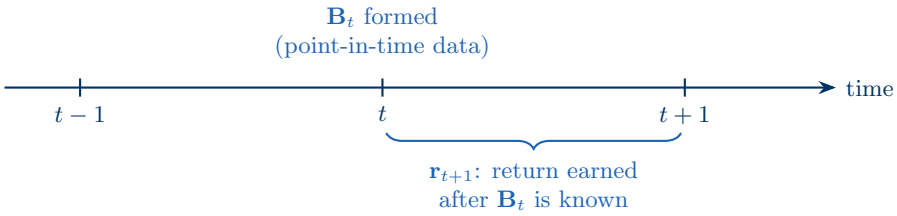
Because $\mathbf{B}$ is known, this is a **cross-sectional** regression: at each date we regress the n asset returns on the n rows of exposures to recover the K factor returns for that date.

2 Timing: Exposures Known Before the Return

The exposures must be measured *before* the return they are used to explain. Writing out the time indices,

$$\mathbf{r}_{t+1} = \mathbf{B}_t\,\mathbf{f}_{t+1} + \boldsymbol{\epsilon}_{t+1}, \tag{2}$$

where $\mathbf{B}_t$ contains only information available at the close of day t, and $\mathbf{r}_{t+1}$ is the return from the close of t to the close of $t+1$.



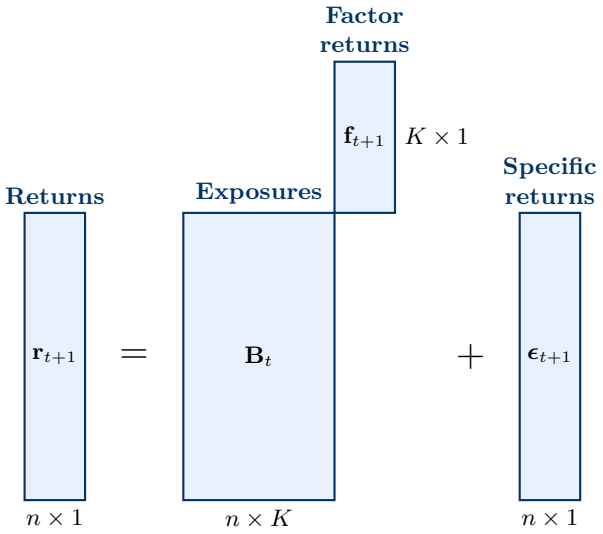
Why not concurrent returns? Many fundamental exposures contain the price: E/P, B/M, dividend yield, market cap, momentum. Regressing $\mathbf{r}_t$ on $\mathbf{B}_t$ built from the close at t creates a mechanical correlation — a stock that rallied today has a lower B/M and a larger size today — so the estimated value and size returns pick up today's price move rather than a premium. It is also look-ahead bias: the resulting factor returns could not have been earned.

"Known at t" means **point-in-time**: accounting data enters $\mathbf{B}_t$ only after its filing or announcement date (e.g. Compustat `rdq`), not at the fiscal period end. For a longer horizon h, regress $\mathbf{r}_{t+1:t+h}$ on $\mathbf{B}_t$.

3 The Cross-Sectional Regression

3.1 Matrix Representation

At each date the regression has the following shape. The n dimension is tall because the universe has many more assets than factors ($n \gg K$); the K columns of $\mathbf{B}_t$ match the K rows of $\mathbf{f}_{t+1}$.



Note: the regression is run separately for each date $t = 1, \ldots, T$, with the exposures $\mathbf{B}_t$ re-formed every period.

3.2 Estimation

With a diagonal weight matrix $\mathbf{W}_t$, the weighted least-squares estimate of the factor returns is

$$\hat{\mathbf{f}}_{t+1} = \left(\mathbf{B}_t^\top \mathbf{W}_t \mathbf{B}_t\right)^{-1} \mathbf{B}_t^\top \mathbf{W}_t \, \mathbf{r}_{t+1}. \tag{3}$$

OLS is the special case $\mathbf{W}_t = \mathbf{I}$. Because idiosyncratic variance differs across stocks, practitioners usually weight by $w_i \propto 1/\sigma_{\epsilon,i}^2$ or, as a proxy, by $\sqrt{\text{market cap}_i}$.

Two practical conventions:

- **Standardize exposures** cross-sectionally at each date (z-scores, winsorized), so that factor returns are comparable over time and across factors.
- **Include an intercept** (a column of ones in $\mathbf{B}_t$, the "market" factor) or a full set of industry dummies, which then play the role of the intercept.

4 From Factor Returns to Risk Premia

Running the regression at every date gives a time series $\hat{\mathbf{f}}_2, \ldots, \hat{\mathbf{f}}_{T+1}$ of estimated factor returns. The **risk premium** of each factor is the expected value of its factor return, estimated by the time-series mean (the Fama–MacBeth procedure):

$$\hat{\boldsymbol{\lambda}} = \frac{1}{T} \sum_{t=1}^{T} \hat{\mathbf{f}}_{t+1}, \qquad \text{SE}(\hat{\lambda}_k) = \frac{\text{sd}(\hat{f}_k)}{\sqrt{T}}. \tag{4}$$

The Fama–MacBeth standard error is robust to cross-sectional correlation in the residuals, because each $\hat{\mathbf{f}}_{t+1}$ already aggregates the whole cross-section. For overlapping multi-day horizons, use Newey–West standard errors with about h lags.

Step	What happens
1. Form exposures	Build $\mathbf{B}_t$ from point-in-time data at the close of t; standardize.
2. Cross-section	Regress $\mathbf{r}_{t+1}$ on $\mathbf{B}_t$ to get $\hat{\mathbf{f}}_{t+1}$ ($K \times 1$).
3. Repeat	Roll forward one period and repeat for $t = 1, \ldots, T$.
4. Average	$\hat{\boldsymbol{\lambda}}$ = time-series mean of $\hat{\mathbf{f}}_{t+1}$; t-stat = $\hat{\lambda}_k / \text{SE}(\hat{\lambda}_k)$.